Jeonghyun Woo

Ph.D. Candidate, Dept. of Electrical & Computer Engineering
The University of British Columbia (UBC)
jhwoo36@ece.ubc.ca | +1 (778) 929-7902 | Website | GitHub | LinkedIn | Google Scholar

RESEARCH INTERESTS

My research advances **computer architecture**, **hardware/systems security**, **memory systems**, and **AI/ML systems**, focusing on:

– **Trustworthy AI Systems:** Designing lightweight cross-layer defenses against hardware-induced privacy and integrity threats in LLM serving, including bit-flip jailbreaks and side-channel leakage from MoE and speculative decoding.

– **Secure and Efficient Memory Systems:** Building high-performance, energy-efficient, and secure memory systems across conventional and emerging technologies, with defenses against RowHammer, denial-of-service, and side-channel attacks.

– **Efficient LLM Inference:** Co-designing algorithms and memory systems to reduce data movement and latency in large-scale LLM serving, including efficient KV-cache management and speculative decoding.

EDUCATION

The University of British Columbia (UBC) — September 2022 – November 2026 (Expected)
Ph.D., Electrical and Computer Engineering | GPA: 4.00/4.00 — *Vancouver, BC, Canada*
Advisor: Prof. Prashant Nair
Dissertation: *Secure and Low-Overhead RowHammer Mitigations for Scalable Memory Systems*

Hanyang University (HYU) — March 2018 – February 2020
M.S., Electronics and Computer Engineering | GPA: 4.00/4.00 — *Seoul, South Korea*
Advisor: Prof. Ki-Seok Chung
Thesis: *RowHammer Mitigation Architecture for Highly Reliable DRAM*

Hanyang University (HYU) — March 2012 – February 2018
B.S., Electronic Engineering, Summa Cum Laude | GPA: 3.89/4.00 — *Seoul, South Korea*

PROFESSIONAL EXPERIENCE

The University of British Columbia (UBC) — September 2022 – Present
*Graduate Research Assistant, **STAR Lab*** — *Vancouver, BC, Canada*
Advisor: Prof. Prashant Nair

NVIDIA Research — May 2024 – August 2024
*Research Intern, **Architecture Research Group*** — *Westford, MA, USA*
Mentor: Dr. Aamer Jaleel | Manager: Dr. David Nellans

Micron Technology — May 2023 – August 2023
*Systems Research Engineering Intern, **Vertical Systems Research*** — *Folsom, CA, USA*
Mentor: Dr. Chun-Yi Liu | Manager: Ameen Akel

Hanyang University (HYU) — March 2018 – February 2020
*Graduate Research Assistant, **ESoC Lab*** — *Seoul, South Korea*
Advisor: Prof. Ki-Seok Chung

HONORS AND AWARDS

Distinguished Artifact Award, HPCA 2025 — *1 of 2 Distinguished Artifacts*	2025
Best Paper Award, HPCA 2023 — *1 of 2 Best Papers from 364 submissions*	2023
Student Travel Grants — ISCA (2025, 2026); HPCA (2023, 2025)	2023, 2025, 2026
Faculty of Applied Science Graduate Award, UBC — $24,800 CAD total	2022, 2023, 2024, 2025
Hanyang Graduate School Scholarship — 70% tuition coverage (~$9,200 CAD/yr)	2018, 2019
Outstanding Paper Award, KICS Fall Conference	2017
Hanyang Academic Excellence Award — Top 1% of ~15,000 university students	2016, 2017
Hanyang Alumni Association Scholarship — Full tuition (~$10,000 CAD/yr)	2016, 2017
Excellent Tutor Award, Engineering Mathematics Tutoring Program, Hanyang University	2016
Hanyang University Scholarship — Full tuition (~$10,000 CAD/yr)	2012, 2013

PUBLICATIONS

Preprints and Under Review

[P1] "PrisonBreak: Jailbreaking Large Language Models with at Most Twenty-Five Targeted Bit-flips."
Zachary Coalson, **Jeonghyun Woo**, Chris S. Lin, Joyce Qu, Yu Sun, Shiyang Chen, Lishan Yang, Gururaj Saileshwar, Prashant J. Nair, Bo Fang, and Sanghyun Hong.
Submitted to *40th Annual Conference on Neural Information Processing Systems* ***(NeurIPS'26)***, 2026.
arXiv

Peer-Reviewed Conference Publications

[C6] "Loaded Dice: Solving the Non-Selection Problem for Scalable Probabilistic RowHammer Defense."
Jeonghyun Woo*, Junsu Kim*, Aamer Jaleel, and Prashant J. Nair. *(*equal contribution)*
53rd Annual International Symposium on Computer Architecture ***(ISCA'26)***, June 2026. *Acceptance Rate: 19.1%.*
Paper Code Slides

[C5] "When Mitigations Backfire: Timing Channel Attacks and Defense for PRAC-Based RowHammer Mitigations."
Jeonghyun Woo, Joyce Qu, Gururaj Saileshwar, and Prashant J. Nair.
52nd Annual International Symposium on Computer Architecture ***(ISCA'25)***, June 2025. *Acceptance Rate: 23.1%.*
Paper Code Slides
[Artifact Evaluated: Available, Functional, Reproduced]

[C4] "QPRAC: Towards Secure and Practical PRAC-Based Rowhammer Mitigation Using Priority Queues."
Jeonghyun Woo, Shaopeng (Chris) Lin, Prashant J. Nair, Aamer Jaleel, and Gururaj Saileshwar.
31st International Symposium on High-Performance Computer Architecture ***(HPCA'25)***, March 2025. *Acceptance Rate: 21.0%.*
Paper Code Slides
[Distinguished Artifact Award — 1 of 2 Distinguished Artifacts]
[Artifact Evaluated: Available, Functional, Reproduced]

[C3] "DAPPER: A Performance-Attack-Resilient Tracker for RowHammer Defense."
Jeonghyun Woo and Prashant J. Nair.
31st International Symposium on High-Performance Computer Architecture ***(HPCA'25)***, March 2025. *Acceptance Rate: 21.0%.*
Paper Slides

[C2] "Scalable and Secure Row-Swap: Efficient and Safe Row Hammer Mitigation in Memory Systems."
Jeonghyun Woo, Gururaj Saileshwar, and Prashant J. Nair.
29th International Symposium on High-Performance Computer Architecture ***(HPCA'23)***, February 2023. *Acceptance Rate: 25.0%.*
Paper Code Slides
[Best Paper Award — 1 of 2 Best Papers from 364 submissions]
[Artifact Evaluated: Available, Functional, Reproduced]

[C1] "HammerFilter: Robust Protection and Low Hardware Overhead Method for RowHammer."
Kwangrae Kim, **Jeonghyun Woo**, Junsu Kim, and Ki-Seok Chung.
39th International Conference on Computer Design ***(ICCD'21)***, October 2021. *Acceptance Rate: 24.4%.*
Paper Slides Video

Workshop Publications and Posters

[W3] "CnC-PRAC: Coalesce, Not Cache, Per-Row Activation Counts for an Efficient in-DRAM Rowhammer Mitigation."
Chris S. Lin, **Jeonghyun Woo**, Prashant J. Nair, and Gururaj Saileshwar.
Fifth Workshop on DRAM Security ***(DRAMSec'25)***, co-located with ISCA 2025, June 2025.
Paper

[W2] "Counterpoint: One-Hot Counting for PRAC-Based RowHammer Mitigation."
Shih-Lien Lu, **Jeonghyun Woo**, and Prashant J. Nair.
Fifth Workshop on DRAM Security ***(DRAMSec'25)***, co-located with ISCA 2025, June 2025.
Paper

[W1] "HammerFilter: Robust Protection and Low Hardware Overhead Method for Row-Hammering."
Kwangrae Kim, Junsu Kim, **Jeonghyun Woo**, and Ki-Seok Chung.
Work-in-Progress Poster, *58th Design Automation Conference* ***(DAC'21)***, December 2021.
Poster

Domestic (Korean) Conference Publications

[**D2**] "A Method to Find the Optimal Probability for Probability-Driven Additional Row Refresh to Prevent DRAM Row Hammering."
Jeonghyun Woo and Ki-Seok Chung.
Korean Institute of Communications and Information Sciences (KICS) Winter Conference, January 2019.

[**D1**] "Implementation of an FPGA-Based CNN Accelerator Using SDSoC."
Changwoo Lee*, **Jeonghyun Woo***, Sang-Soo Park, and Ki-Seok Chung. *(*equal contribution)*
KICS Fall Conference, November 2017.
Code **[300+ GitHub stars]**
[Outstanding Paper Award]

INVITED TALKS

When Optimization Breaks Security: Architecting Efficient and Trustworthy Hardware
Nanyang Technological University, Singapore, July 2026

Built on Broken Ground: Why Secure Computing Must Start at the Hardware
University of Waterloo, Waterloo, Canada, March 2026

Towards Secure and Scalable Memory Systems: From RowHammer Attacks to Hardware-Induced LLM Jailbreaks
KAIST CASYS Seminar, Remote, November 2025

Towards Secure and Low-Overhead PRAC-Based RowHammer Mitigations
NVIDIA Research (end-of-internship talk), Westford, USA, August 2024

RowHammer Mitigations for Future High-Bandwidth Memory
Micron Technology (end-of-internship talk), Folsom, USA, August 2023

Conference Talks

Loaded Dice: Solving the Non-Selection Problem for Scalable Probabilistic RowHammer Defense
ISCA'26, Raleigh, USA, June 2026

When Mitigations Backfire: Timing Channel Attacks and Defense for PRAC-Based RowHammer Mitigations
ISCA'25, Tokyo, Japan, June 2025

QPRAC: Towards Secure and Practical PRAC-Based Rowhammer Mitigation Using Priority Queues
HPCA'25, Las Vegas, USA, March 2025

DAPPER: A Performance-Attack-Resilient Tracker for RowHammer Defense
HPCA'25, Las Vegas, USA, March 2025

Scalable and Secure Row-Swap: Efficient and Safe Row Hammer Mitigation in Memory Systems
HPCA'23, Montreal, Canada, February 2023

Implementation of an FPGA-Based CNN Accelerator Using SDSoC
KICS'17 Fall, Daegu, South Korea, November 2017

PROFESSIONAL SERVICE

Program Committee

- 2026 IEEE/ACM International Symposium on Microarchitecture (**MICRO'26**) — Student PC
- 2026 IEEE International Symposium on High-Performance Computer Architecture (**HPCA'26**) — Light PC

Artifact Evaluation Committee

- 2025 IEEE International Symposium on Workload Characterization (**IISWC'25**)

Invited Reviewer

- IEEE Transactions on Very Large Scale Integration Systems (**TVLSI**), 2025

Student Volunteer

- 2025 SPICE Workshop, co-located with MICRO'25
- 2024 IEEE International Symposium on Workload Characterization (**IISWC'24**)
- 2023 International Conference on Architectural Support for Programming Languages and Operating Systems (**ASPLOS'23**)

TEACHING EXPERIENCE

The University of British Columbia (UBC) Vancouver, BC, Canada
Graduate Teaching Assistant
CPEN 411: Computer Architecture 2022, 2023, 2024
CPEN 312: Digital Systems and Microcomputers 2025

Hanyang University (HYU) Seoul, South Korea
Graduate Teaching Assistant
ELE 3081: VLSI Engineering 2019
ITE 4003: SoC Design 2018

MENTORING EXPERIENCE

Junsu Kim January 2020 – Present
Ph.D. Student, The University of British Columbia (UBC) *Vancouver, BC, Canada*
Project: Scalable and robust RowHammer mitigations → **Published at:** DAC'21 (WIP), ICCD'21, ISCA'26
Project: Efficient LLM inference via KV-cache management and speculative decoding

Kwangrae Kim January 2019 – December 2021
Ph.D. Student, Hanyang University (HYU) *Seoul, South Korea*
Project: Low-cost and robust probabilistic RowHammer mitigation → **Published at:** DAC'21 (WIP), ICCD'21

SELECTED ACADEMIC PROJECTS

Compression-Assisted Last-Level Cache Replacement January 2023 – April 2023
Advanced Computer Architecture, UBC (Instructor: Prof. Mieszko Lis)
Showed that 72.7% of cache lines are compressible by $\geq$10 B; proposed a compression-assisted LLC replacement policy that improves hit rates with minimal metadata overhead.

Revisiting Address Translation on Intel Optane DC PMEM September 2022 – December 2022
Graduate Operating Systems, UBC (Instructor: Prof. Margo Seltzer)
Quantified address-translation overheads on Optane DC PMEM with graph processing, HPC, and genomics workloads; demonstrated that Transparent Huge Pages mitigate 4 KB page bottlenecks.

Modified LeNet-5 Forward Pass in CUDA November 2020 – December 2020
Applied Parallel Programming, UIUC
Implemented optimized convolutional forward passes using shared-memory tiling, constant memory, and loop unrolling; profiled with Nsight Systems and Nsight Compute.

TECHNICAL SKILLS

Programming: C/C++, Python, CUDA, Verilog, Perl, Bash, Go, Assembly
Simulators: Ramulator, ChampSim, gem5, DRAMSim2, GPGPU-Sim, MGPUSim
Hardware Tools: Pin, SimPoint, Xilinx Vivado, Xilinx SDSoC, Intel Quartus